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To cite this article: T. R. Buckley , R. L. Palma , P. M. Johns , D. M. Gleeson , A. C.G. Heath , R. A. Hitchmough & I. A.N. Stringer (2012) The conservation status of small or less well known groups of New Zealand terrestrial invertebrates, *New Zealand Entomologist*, 35:2, 137-143, DOI: [10.1080/00779962.2012.686319](https://doi.org/10.1080/00779962.2012.686319)

To link to this article: <http://dx.doi.org/10.1080/00779962.2012.686319>



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The conservation status of small or less well known groups of New Zealand terrestrial invertebrates

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A total of 27 taxa from taxonomic groups with few species, or that are less well known, are listed as THREATENED, 50 taxa are AT RISK, 110 taxa are DATA DEFICIENT and three taxa are EXTINCT. Thirteen taxa are NATIONALLY CRITICAL: *Aceria clianthi*; *Eriophyoidea incertae* (Acari); *Cryptops* sp.; *Haasiella* sp. (Chilopoda); *Burmjapyx* sp. (Diplura); *Hirudobdella antipodum* (Hirudinea); *Antipone-mertes allisonae* (Nemertini); *Prasmiola unica* (Opiliones); and *Tepakiphasma ngatikuri* (Phasmatodea) are freelifving whereas the lice *Apterygon okarito*, *Coloceras harrisoni*, *Rallicola takahe* and *Saemundsson-ia chathamensis* (Phthiraptera) have the same threat status as their bird hosts. No taxa were considered NATIONALLY ENDANGERED but 14 ectoparasites are NATIONALLY VULNERABLE, including six Acari and eight Phthiraptera. The AT RISK taxa comprise two that are DECLINING, four that are RECOVERING, one that is RELICT and 45 taxa that are NATURALLY UNCOMMON. Earthworms (Oligochaeta) also make up 101 of the 110 DATA DEFICIENT taxa. All of the EXTINCT species were host-specific feather lice: two were on extinct birds and one became extinct when its host was transferred to predator-free islands. Thirty-six earthworm species that were previously DATA DEFICIENT are now ranked NOT THREATENED, as are five Phthiraptera that were previously ranked either NATIONALLY CRITICAL or NATIONALLY ENDANGERED and one Phasmatodea that was previously ranked NATIONALLY ENDANGERED.

Keywords: Acari, Oligochaeta, Phasmatodea, Phthiraptera, threatened species

Introduction

The four previous lists of New Zealand threatened species included invertebrate groups with few species or that are less well known. Two taxa, the onychophoran *Peripatoides indigo* (= *Ooperipatellus indigo* in Molloy & Davis 1992) and the leech *Hirudobdella antipodum*, were included in the first published list of threat classifications for New Zealand taxa (Table 1). Both were category 'I', species with insufficient information known but which were likely to be threatened (Molloy & Davis 1992). Only *H. antipodum* was included in the second published list and it

was upgraded to category 'A' the highest priority species for conservation management (Molloy et al. 1994), and another 10 taxa were further added to the category 'I' (Molloy et al. 1994). These comprised a terrestrial flatworm (*Artio-posthia mariae*), three earthworms (*Octochaetus multiporus*, *O. michaelseni*, *O. antarcticus*), an opilione (*Zeopsopilio neozealandiae*), a terrestrial amphipod (*Tara taranaki*), two centipedes (indeterminate taxa of *Haasiella* and *Cryptops*), an onychophoran (*Ooperipatellus viridimaculatus*) and a dipluran (an indeterminate taxa of *Burmjapyx*).

Table 1. Numbers of species included in previous lists of THREATENED New Zealand invertebrates from small groups or poorly studied groups from Molloy and Davis (1992), Molloy et al. (1994), Hitchmough (2002), Hitchmough et al. (2007).

Category	1992	1994	Category	2002	2007
Highest priority spp.	0	1	NATIONALLY CRITICAL	12	14
Second priority spp.	0	0	NATIONALLY ENDANGERED	12	12
Third priority spp.	0	0	NATIONALLY VULNERABLE	0	1
			SERIOUS DECLINE	1	1
			GRADUAL DECLINE	4	4
			SPARSE	0	0
			RANGE RESTRICTED	10	12
Threatened, but few data	2	10	DATA DEFICIENT	168	170
Extinct	0	0		0	1
Total	2	11		207	215

Details about the conservation requirements for the species listed by Molloy et al. (1994) were published by McGuinness (2001). More taxa were added in the next two published lists (Hitchmough 2002; Hitchmough et al. 2007; see Supplementary Appendix 1 (Small Groups)) partly because of new information and partly because of changes in the criteria used for classifying threats. However, the flatworm, opilione, onychophoran and one of the earthworms (*O. multiporus*) were no longer considered THREATENED.

This paper presents the threat classifications of New Zealand invertebrate taxa in Classes or Orders containing relatively few species or in groups containing taxa that have been poorly studied. It includes ticks and some mites (Acari), centipedes (Chilopoda), harvestmen (Opiliones), peripatus (Onychophora), terrestrial flatworms (Platyhelminthes), terrestrial proboscis worms (Nemertini), terrestrial annelids (Annelida: Oligochaeta and Hirudinea), Diplura and the insect orders containing dragonflies (Odonata), earwigs (Dermaptera), parasitic lice (Phthiraptera) and stick insects (Phasmatodea). The threat classifications for invertebrate taxa in speciose groups are presented elsewhere (Andrew et al. 2012; Leschen et al. 2012; Mahlfeld et al. 2012; Sirvid et al. 2012; Stringer et al. 2012a; Stringer et al. 2012b; Trewick et al. 2012; Ward et al. 2012; Yeates et al. 2012). We follow a recent revision by the New Zealand Department of Conservation (DOC) of the criteria used to classify threatened species (Townsend et al.

2008). A summary of the criteria used and changes that were made to the criteria since the last threat lists of Hitchmough et al. (2007) is provided by Stringer and Hitchmough (2012) with full explanations given by Townsend et al. (2008) and de Lange et al. (2009).

Methods

The assessment process was undertaken by a panel of experts with extensive experience with the various taxa (the authors of this paper). Most panel members met on 29 April 2010 and all subsequently communicated with I.A.N. Stringer and R.A. Hitchmough to place taxa into threat categories. The role of this panel and details of the method follow Stringer and Hitchmough (2012), including the opportunity for public to make submissions. However, no public submissions on any of the groups were received. No attempt was made to assess the status of the entire known fauna of the groups considered; the panel members concentrated on taxa listed previously and taxa for which scarcity of material in collections or the experience of the panel suggested that they were likely to be uncommon. This process was guided by panel knowledge and reference to recent publications relating to taxonomic and population status. Where there was doubt in the placement of a given taxon into a threat category, we referred our provisional assessments to other relevant experts subsequent to the workshop.

The main abbreviations used are: THREATENED: NC = NATIONALLY CRITICAL; NE = NATIONALLY ENDANGERED; NV = NATIONALLY VULNERABLE. AT RISK: Dc = DECLINING; Rc = RECOVERING; RI = RELICT; NU = NATURALLY UNCOMMON. Other categories: DD = DATA DEFICIENT; Ex = EXTINCT. For full definitions of abbreviations and terminology see Townsend et al. (2008).

Results

Overall 190 taxa were included, with 27 taxa classified as THREATENED (13 are NATIONALLY CRITICAL and 14 NATIONALLY VULNERABLE), and 50 taxa as AT RISK, most of which are NATURALLY UNCOMMON with two that are DECLINING and four that are RECOVERING (Table 2). A further 110 taxa, mostly earthworms, are DATA DEFICIENT and three taxa are EXTINCT. A complete list of all taxa and their threat status can be found in Supplementary Appendix 2 (Small Groups).

Acari

All of the acarines that are listed as THREATENED either form galls on plants or are parasitic on kiwi (*Apteryx* spp.), tuatara (*Sphenodon punctatus*) or athoracophorid slugs. Both of the gall-forming mites are NATIONALLY CRITICAL. One,

Aceria clianthi (Eriophyidae), has been recorded only from kakabeak (*Clianthus* spp.) in cultivation and once on *Lotus cornalatus* (Fabaceae), an introduced plant growing near kakabeak (Martin 2009). It is given the same threat classification as kakabeak (de Lange et al. 2009). The second is an unidentified mite that forms leaf-pocket galls on *Carpodetus serratus*. It is known only from a single tree at Nelson Lakes despite extensive searching throughout New Zealand (N. Martin, personal communication). The six NATIONALLY VULNERABLE taxa are all host specific on various kiwi taxa and include four feather mites, a tick and the immature stages of a trombiculid mite (Heath 2010a). AT RISK Acari comprise a single species of tick, *Amblyomma sphenodonti*, which is host specific on tuatara and ranked RELICT, whereas two other mite species, *Riccardoella novaezealandiae* and another indeterminate *Riccardoella*, which occur in the pallial cavities of athoracophorid slugs, are ranked DATA DEFICIENT (Fain & Barker 2004; G. Barker, personal communication).

Oligochaeta

All the listed earthworms are classified either as NATURALLY UNCOMMON (30 taxa) or DATA DEFICIENT (102 taxa). The NATURALLY UNCOMMON species are ISLAND ENDEMICS known from single

Table 2. Threat status of small or less well known groups of terrestrial invertebrates in New Zealand. Data follow the criteria of Townsend et al. (2008).

	THREATENED			AT RISK			Others			Total
	NC	NE	NV	Dc	Rc	RI	NU	DD	Ex	
Acari	2		6			1		2		11
Amphipoda							1			1
Chilopoda	2									2
Diplura	1									1
Hirudinea	1						1			2
Mantodea				1						1
Nemertini	1									1
Oligochaeta							30	102		132
Onychophora							1	1		2
Opiliones	1									1
Phasmatodea	1						3	1		5
Phthiraptera	4		8	1	4		7	3	3	30
Platyhelminthes								1		1
Total	13	0	14	2	4	1	43	110	3	190

Note: See Methods for abbreviations.

islands except for *Plutellus aucklandicus* which occurs on both of the largest Auckland Islands (Lee 1959).

Phthiraptera

Twelve louse taxa are THREATENED, comprising four species that are NATIONALLY CRITICAL and eight taxa that are NATIONALLY VULNERABLE. A further 12 are AT RISK: one is DECLINING; seven are NATURALLY UNCOMMON; and four are RECOVERING. Three taxa are DATA DEFICIENT and three species are EXTINCT. The high numbers of Phthiraptera (see Supplementary Appendix 1 (Small Groups)) resulted from these host-specific parasitic lice living on birds which also had these threat classifications. This is because the threat status of parasitic taxa follows that of their hosts unless they are clearly rarer than their hosts in which case they deserve a higher threat status.

Other groups

We also listed the following terrestrial invertebrates as NATIONALLY CRITICAL: two cave centipedes (*Cryptops* sp., *Haasiella* sp.); a leech (*Hirudobdella antipodum*); a cave dipluran (*Burmjapyx* sp.); a terrestrial nemertine (*Antiponemertes allisonae*); and a harvestman (*Prasmiola unica*). Five stick insects, of which one (*Tepakiphasma ngatikuri*) is NATIONALLY CRITICAL, three (including two indeterminate taxa of *Clitarchus* and *Pseudoclitarchus sentus*) are NATURALLY UNCOMMON and one indeterminate taxon of *Micrarchus* is DATA DEFICIENT. Other taxa include a praying mantis (*Orthodera novaezealandiae*) classified as DECLINING, a leech (*Ornithobdella edentula*) that is NATURALLY UNCOMMON, two peripatus (*Ooperipatellus nanus*, NATURALLY UNCOMMON; *Peripatoides indigo*, DATA DEFICIENT), one DATA DEFICIENT terrestrial flatworm (*Arthurdendyus albidus*), and a NATURALLY UNCOMMON land amphipod (*Tara taranaki*).

Lastly, there are three EXTINCT species; all are host-specific feather lice. Two, *Philopteroides xenicus* and *Rallicola (Huiacola) extinctus*, because their respective hosts, the South Island bush wren and the huia, are extinct. The host of *Rallicola (Aptericola) pilgrimi*, the little spotted kiwi, survived only after translocation to predator-free islands but the louse did not survive the relocation.

Discussion

The threat rankings of two mites that were previously listed by Hitchmough et al. (2007) have been changed because of changes made to both the categories and the ranking criteria. The slug mite *R. novaezealandiae* was moved from RANGE RESTRICTED to DATA DEFICIENT even though no additional information has been collected, whereas the tuatara tick, *A. sphenodonti*, was changed from RANGE RESTRICTED to RELICT following the change in threat status of its host (Hitchmough et al. 2010). The other nine acarines are new additions: two form galls and seven are ectoparasitic on kiwi (*Apteryx* species). Of the latter, four occur on more than one kiwi species and they take the ranks of their least threatened host. The last two mites, *Kiwilichus delosikyus* and *Kiwialges haastii*, are known only from South Island brown kiwi and great spotted kiwi respectively (Heath 2010a, 2010b).

No additional species of earthworm have been added since the last threatened species list was published (Hitchmough et al. 2007), but the rankings of 67 have changed. These include 30 previously DATA DEFICIENT species that are now ranked as NATURALLY UNCOMMON because additional information became available: 28 are recorded from single islands and two are recorded from island groups. Another 36, previously listed as DATA DEFICIENT, are now ranked as NOT THREATENED mostly because their known ranges have since been expanded following further research on them in northern New Zealand (Buckley & Bartlam 2010; Buckley et al. 2011). Three of these previously DATA DEFICIENT species are widespread (*Megascolides suteri*, *M. maoricus*, *Diporochoaeta obtusa*) and were incorrectly ranked as DATA DEFICIENT in Hitchmough et al. (2007). Their status has now been changed to NOT THREATENED. Finally, *Deinodrilus montanus* was previously considered NOT THREATENED whereas we now classify it as DATA DEFICIENT. This earthworm has been collected only twice from the same locality near Denniston, but the surrounding region on the West Coast of the South Island has been poorly surveyed for earthworms. Many earthworm species have restricted geographical distributions (e.g., King et al. 2008). Therefore, the restricted geographic ranges reflected in the threat listings here may reflect natural biogeographic processes rather than loss of habitat.

The threat status of many Phthiraptera have changed and only six of the 34 taxa listed by Hitchmough et al. (2007) now retain equivalent threat rankings. This is partly due to changes in the threat rankings of their hosts and partly to a re-appraisal of their host species with the lice subsequently being assigned the threat rankings of their least threatened hosts. This re-appraisal has, for example, resulted in four species and an indeterminate entity being classified as NOT THREATENED. The overall result is that there are now fewer highly THREATENED lice taxa and an increase in less THREATENED lice taxa. There are now four lice taxa listed as NATIONALLY CRITICAL and eight as NATIONALLY VULNERABLE compared with 10 NATIONALLY CRITICAL, 14 NATIONALLY ENDANGERED and one NATIONALLY VULNERABLE in Hitchmough et al. (2007). Conversely, the AT RISK taxa have increased from the eight listed in Hitchmough et al. (2007) to 12 in this review. DATA DEFICIENT taxa have increased from zero to three and there are now three EXTINCT species with the addition of *Rallicola* (*Apterocola*) *pilgrimi* and *Rallicola* (*Huiacola*) *extinctus*. The former was host-specific on little spotted kiwi, which is now locally extinct on the mainland and no lice survived the transfers of this bird to island locations, as mentioned above, whereas the latter was omitted from previous threat lists.

Of the remaining species, three NATIONALLY CRITICAL species (*Antiponemertes allisonae* (Nemertini), *Hirudobdella antipodum* (Hirudinea), *Prasmiola unica* (Opiliones)) and one DATA DEFICIENT species (*Arthurdendyus albidus* (Turbellaria)) have retained their threat status, whereas five formerly RANGE RESTRICTED taxa (*Tara taranaki* (Amphipoda), *Ornithobdella edentula* (Hirudinea), *Ooperipatellus namus* (Onychophora), *Clitarchus* sp. aff. *hookeri* from the Poor Knights and *Pseudoclitarchus sentus* (Phasmatodea)) have been placed in the equivalent new category, NATURALLY UNCOMMON. Two cave centipedes, *Cryptops* sp. and *Haasiella* sp., which were formerly DATA DEFICIENT, are now considered NATIONALLY CRITICAL as is the cave dipluran, *Burmjapyx* sp., which was formerly RANGE RESTRICTED. All are known from single caves and none has been found elsewhere despite further searching. *Peripatoides indigo* (Onychophora) was common in Twin Forks Cave and classified as RANGE RESTRICTED, but it

is now known from two other non-cave localities so its status has changed to DATA DEFICIENT, whereas *Peripatoides suteri* is now thought to be a genetic polymorphism (an extra pair of legs) of *P. novaezealandiae*, a species which is NOT THREATENED.

One species, the stick insect *Asteliaphasma naomi*, has now been found in so many localities that its status has been changed from NATIONALLY CRITICAL to NOT THREATENED. Finally, four new taxa have been added to the threat list comprising three stick insects and a praying mantis. The stick insects include the NATIONALLY CRITICAL *Tepakiphasma ngatikuri*, known from two specimens from the end of the Aupouri Peninsula (Buckley & Bradler 2010), and an indeterminate NATURALLY UNCOMMON species of *Clitarchus* from the same area. Because this area of the Aupouri Peninsula is a known area of endemism (Buckley & Bradler 2010), these two species are likely to be restricted to it. The last stick insect included on the list, as DATA DEFICIENT, is an indeterminate *Micrarchus* from Stephen's Island. Although collected only from this island, it is not known if it is more widespread in the Marlborough Sounds. We consider the broad geographic distributions of all stick insects on the threatened species list are the product of biogeographic history rather than a massive loss of habitat. The distribution of *Tepakiphasma ngatikuri* has probably contracted in the Te Paki area due to deforestation, but introduced mammals and vespulid wasps are also likely threats to this species. The New Zealand praying mantis, *Orthodera novaesealandiae*, was classified as DECLINING because its numbers are thought to be reduced through competition as the introduced African mantis, *Miomantis caffra*, increases its range in the South Island (Ramsay 1990).

Overall, of the 187 taxa we have listed as THREATENED (27), AT RISK (50) and DATA DEFICIENT (110), the threat classifications of 110 remain essentially the same as previously listed by Hitchmough et al. (2007). Fourteen are additions, the status of 58 has changed because new information became available for them, and five changed status primarily because the criteria were changed. Forty two taxa, previously classified as THREATENED or DATA DEFICIENT, have now been classified as NOT THREATENED. We have not covered every group of

terrestrial invertebrate present in New Zealand, but the intention is to increase the range of taxa assessed in subsequent years to eventually cover all species. It should, for example, be possible to cover the fleas (Siphonaptera) as well as more Acari, such as more feather mites and the parasitic mites of lizards and ticks, including the recently described tick from the native short-tailed bat (Heath et al. 2011; Heath in press) when terrestrial invertebrates are next reassessed.

Acknowledgements

We thank the following for providing advice and information: Gregor Yeates for earthworms, Nicholas Martin for gall mites, Gary Barker for slug mites, Dallas Bishop for feather mites, and Graeme Ramsay together with members of the Entomological Society of New Zealand for the native praying mantis. Research on earthworms and stick insects was supported by the New Zealand Ministry of Science and Innovation through the Defining New Zealand's Land Biota programme.

References

- Andrew IG, Macfarlane RP, Johns PM, Hitchmough RA, Stringer IAN. 2012. The conservation status of New Zealand Diptera. *New Zealand Entomologist* 35: 99–102.
- Buckley TR, Bradler S. 2010. *Tepakiphasma ngatikuri*, a new genus and species of stick insect (Phasmatodea) from the Far North of New Zealand. *New Zealand Entomologist* 33: 118–126.
- Buckley TR, Marske K, Attanayake D. 2010. Phylogeography and ecological niche modelling of the New Zealand stick insect *Clitarchus hookeri* (White) support survival in multiple coastal refugia. *Journal of Biogeography* 37: 682–695.
- Buckley TR, Allwood JJ, Bartlam S, Howitt R, Prada D. 2011. Phylogenetic analysis of New Zealand earthworms (Oligochaeta: Megascolecidae) reveals ancient clades and cryptic taxonomic diversity. *Molecular Phylogenetics and Evolution* 58: 85–96.
- de Lange PJ, Norton DA, Courtney AP, Heenan PB, Barkla JW, Cameron EK, Hitchmough R, Townsend AJ. 2009. Threatened and uncommon plants of New Zealand (2008 revision). *New Zealand Journal of Botany* 47: 61–96.
- Fain A, Barker GM. 2004. A new genus and species of mite of the family Erythetidae (Acari Prostigmata) from the pallial cavity of a New Zealand terrestrial gastropod (Athyridae: Athyridae): *Bulletin de la Société Royale Belge d'Entomologie* 140: 43–48.
- Heath ACG. 2010a. A review of ectoparasites of *Apteryx* spp. (kiwi) in New Zealand, with new host records, and the biology of *Ixodes anatis* (Acari: Ixodidae). *Tuhinga* 21: 147–159.
- Heath ACG. 2010b. Checklist of ectoparasites of birds in New Zealand: additions and corrections. *Surveillance* 37: 2–15.
- Heath ACG. In press. A new species of soft tick (Ixodoidea: Argasidae) from the New Zealand lesser short-tailed bat, *Mystacina tuberculata* Gray. *Tuhinga*.
- Heath ACG, Palma RL, Cane RP, Hardwick S. 2011. Checklist of New Zealand ticks (Acari: Ixodidae, Argasidae). *Zootaxa* 2995: 55–63.
- Hitchmough RA. 2002. New Zealand Threat Classification System Lists 2002. Threatened Species Occasional Publication No. 23. Department of Conservation, Wellington. 210 p.
- Hitchmough R, Bull L, Cromarty P. 2007. New Zealand Threat Classification System Lists 2005. Department of Conservation, Wellington. 194 p.
- Hitchmough RA, Hoare JM, Jamieson H, Newman D, Tocher MD, Anderson PJ, Lettink M, Whitaker AH. 2010. Conservation status of New Zealand reptiles, 2009. *New Zealand Journal of Zoology* 37: 203–224.
- King RA, Tibble AL, Symondson WOC. 2008. Opening a can of worms: unprecedented sympatric cryptic diversity within British lumbricid earthworms. *Molecular Ecology* 17: 4684–4698.
- Leschen RAB, Marris JWM, Emberson RM, Nunn J, Hitchmough RA, Stringer IAN. 2012. The conservation status of New Zealand Coleoptera. *New Zealand Entomologist* 35: .
- Lee KE. 1959. The Earthworm Fauna of New Zealand. Bulletin 130. Department of Scientific and Industrial Research, Wellington. 486 p.
- McGuinness C. 2001. The Conservation Requirements of New Zealand's Nationally Threatened Invertebrates. Threatened Species Occasional Publication No. 20, Department of Conservation, Wellington. 657 p.
- Mahlfeld K, Brook FJ, Roscoe DJ, Hitchmough RA, Stringer IAN. 2012. The conservation status of New Zealand terrestrial Gastropoda excluding *Powelliphanta*. *New Zealand Entomologist* 35: .
- Martin NA. 2009. Herbivores and pathogens associated with *Clanthus* species (Fabales: Fabaceae). *The Weta* 38: 6–12.
- Molloy J, Davis A. 1992. Setting Priorities for the Conservation of New Zealand's Threatened Plants and Animals. Department of Conservation, Wellington. 44 p.
- Molloy J, Davis A, Tisdall C. 1994. Setting Priorities for the Conservation of New Zealand's Threatened Plants and Animals. 2nd ed. Department of Conservation, Wellington. 64 p.
- Ramsay GW. 1990. Mantodea (Insecta), With a Review of Aspects of Functional Morphology and Biology. Fauna of New Zealand 19. DSIR Publishing, Wellington. 96 p.

- Sirvid PJ, Vink CJ, Wakelin MD, Fitzgerald BM, Hitchmough RA, Stringer IAN. 2012. The conservation status of New Zealand Araneae. *New Zealand Entomologist* 35: 85–90.
- Stringer IAN, Hitchmough RA. 2012. Assessing the conservation status of New Zealand's native terrestrial invertebrates. *New Zealand Entomologist* 35: 77–84.
- Stringer IAN, Hitchmough RA, Larivière M-C, Eyles AC, Teulon DAG, Dale PJ, Henderson RC. 2012a. The conservation status of New Zealand Hemiptera. *New Zealand Entomologist* 35: 100–115.
- Stringer IAN, Hitchmough RA, Dugdale JS, Edwards E, Hoare RJB, Patrick BH. 2012b. The conservation status of New Zealand Lepidoptera. *New Zealand Entomologist* 35: 120–127.
- Townsend AJ, de Lange PJ, Duffy CAJ, Miskelly CM, Molloy J, Norton DA. 2008. *New Zealand Threat Classification System Manual*. Department of Conservation, Wellington. 35 p.
- Trewick SA, Morris SJ, Johns PM, Hitchmough RA, Stringer IAN. 2012. The conservation status of New Zealand Orthoptera. *New Zealand Entomologist* 35: 131–136.
- Ward DF, Early JW, Schnitzler F-R, Hitchmough RA, Stringer IAN. 2012. The conservation status of New Zealand Hymenoptera. *New Zealand Entomologist* 35: 116–119.
- Yeates GW, Zhao ZQ, Hitchmough RA, Stringer IAN. 2012. The conservation status of New Zealand Nematoda. *New Zealand Entomologist* 35: 128–130.